Práctica Dos: Sistema Cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

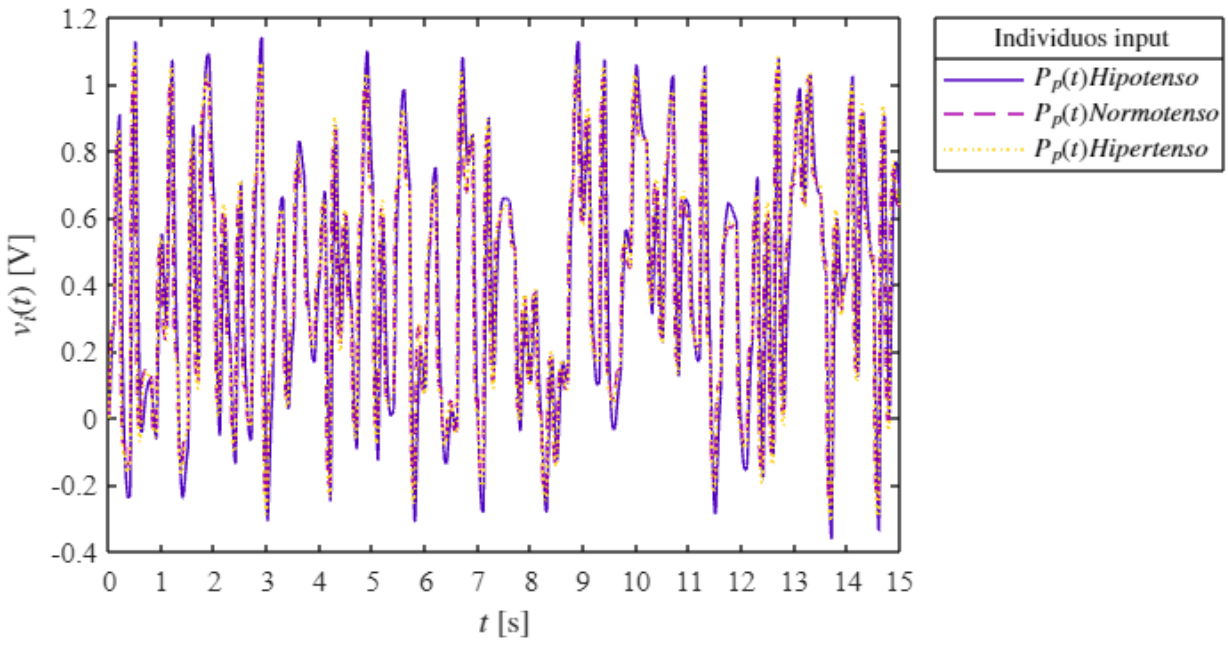
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Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system("SistemaP2")
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';
x = sim('SistemaP2', parameters);
t_data= (0:0.1:15);
t=(0:0.001:15);
Pa=interp1(t_data, x.PA, t, 'previous');
writematrix(x.PA,'signal.xlsx');
```

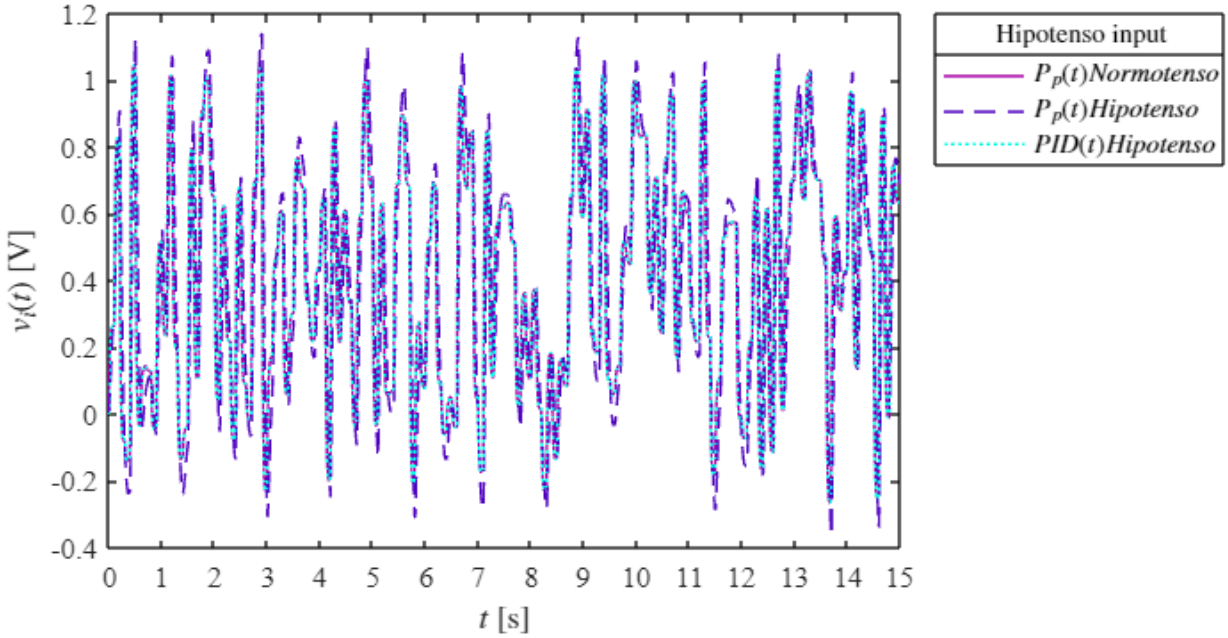
Respuestas: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t,x.PpL,x.PpN,x.PpH,'Individuos ');
```



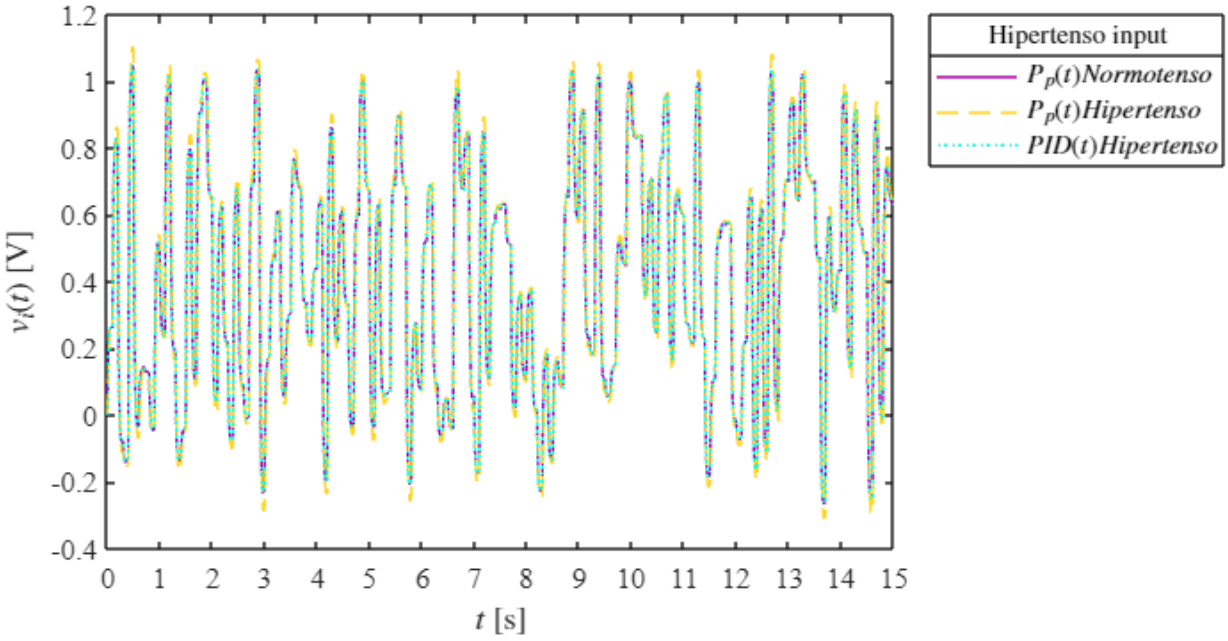
Respuestas: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'Hipotenso ');
```



Respuestas: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpH,x.PIDH,'Hipertenso ');
```

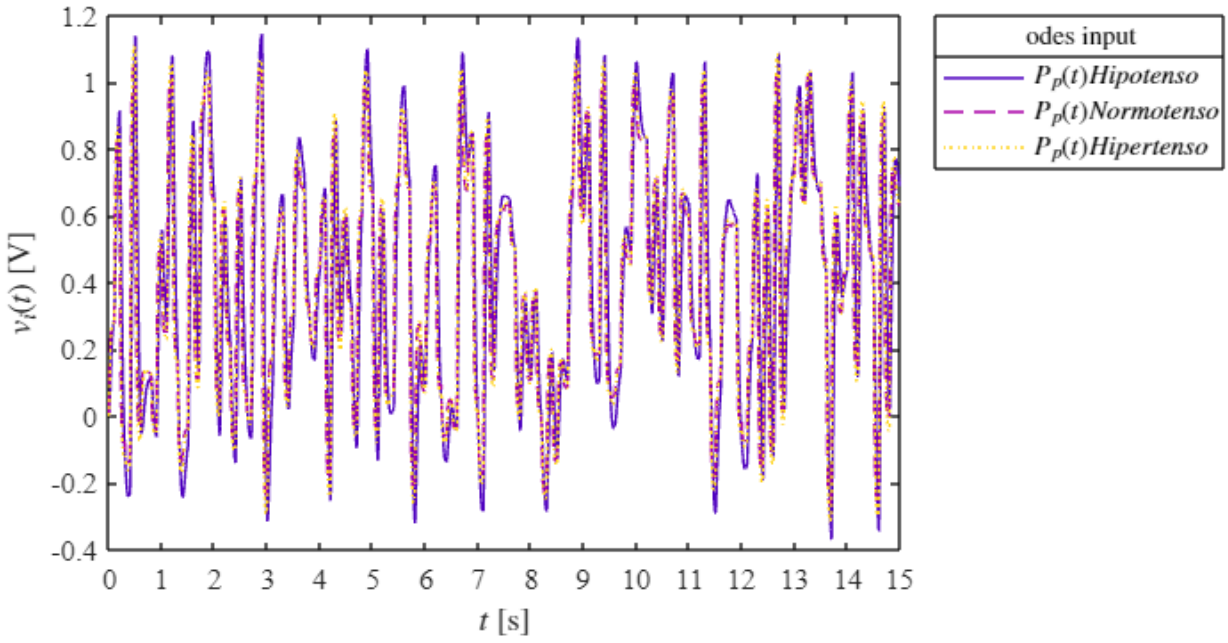


Modelo de EDOs

```
par_hipo = [0.01,0.001,1.00,3.500];
par_normo = [0.01,0.001,1.00,1.500];
par_hiper = [0.01,0.001,1.00,1.000];

PpN = cardio(par_normo, Pa, t);
PpL = cardio(par_hipo, PpN, t);
PpH = cardio(par_hiper, PpN, t);

plotsignals(t,PpL,PpN,PpH,'odes ');
```




```
function Pp = cardio(par,Pa,t)
    Z= par(1); L = par(2); R = par(3); C= par(4); dt=t(2);
    n = round(t(end)/dt);
    FL = zeros(length(t),1); FL(1)=0;
    Pp = zeros(length(t),1); Pp(1)=0;
    for i = 1:n
        FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;
    end

end
```

Función: Respuestas de la simulación

```
function plotsignals(t,x1,x2,x3,caso)
set(figure(), "color", 'w')
set(gcf, 'units', 'inches','position',[1,1,7,3.5])
set(gca, 'fontname', 'times new roman', 'fontsize', 11)
```

```
hold on; grid off, box on;
clr1 = [80,3,192]/255;
clr2 = [171,3,169]/255;
clr3 = [255,213,30]/255;
clr4 = [3,252,232]/255;
order = [clr1;clr2;clr3:clr4]; colororder(order)

if caso == "Individuos "

plot(t,x1,'-','LineWidth',0.25,'color',clr1) %hipo
plot(t,x2,'--','LineWidth',0.25,'Color',clr2) %normo
plot(t,x3,':','LineWidth',0.25,'Color',clr3) %hiper
L = legend('$P_{p}(t) Hipotenso$','$P_p(t) Normotenso$','$P_p(t) Hipertenso$');

elseif caso =="Hipotenso "
    plot(t,x1,'-','LineWidth',0.25,'color',clr2) %normo
plot(t,x2,'--','LineWidth',0.25,'Color',clr1) %hipo
plot(t,x3,':','LineWidth',1,'Color',clr4) %controlador del hipo
L = legend('$P_{p}(t) Normotenso$','$P_p(t) Hipotenso$','$PID(t) Hipotenso$');

elseif caso == "Hipertenso "
    plot(t,x1,'-','LineWidth',0.25,'color',clr2) %normo
plot(t,x2,'--','LineWidth',0.25,'Color',clr3) %hiper
plot(t,x3,':','LineWidth',1,'Color',clr4) %hControlador del hiper
L = legend('$P_{p}(t) Normotenso$','$P_p(t) Hipertenso$','$PID(t) Hipertenso$');

elseif caso == "odes "
    plot(t,x1,'-','LineWidth',0.25,'color',clr1) %hipo
    plot(t,x2,'--','LineWidth',0.25,'color',clr2) %normo
    plot(t,x3,':','LineWidth',0.25,'color',clr3) %hiper
    L = legend('$P_{p}(t) Hipotenso$','$P_p(t) Normotenso$','$P_p(t) Hipertenso$');
end

set(L,'interpreter','latex','location','northeastoutside','orientation','vertical')
title(L,[caso,'input'])
xlabel('$t$ [s]','interpreter','latex')
ylabel('$v_i(t)$ [V]','Interpreter','latex')
xlim([0 15]); xticks(0:1:15)
ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

exportgraphics(gcf,[caso,'_matlab.pdf'],'ContentType','vector')
end
```